

The Impact of Socioeconomic Status on Maternal Healthcare Utilization in Nepal: Evidence from the Nepal Demographic Health Survey 2016



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Abstract

This study examines how socioeconomic status—specifically household wealth and maternal education—affects the utilization of maternal healthcare services in Nepal. Using cross-sectional data from the 2016 Nepal Demographic and Health Survey (NDHS), four key outcomes are analyzed: institutional delivery, number of antenatal care (ANC) visits, postnatal care uptake, and infant mortality. Multivariate logistic and OLS regressions indicate that higher wealth and education are strongly associated with increased use of maternal health services and improved infant survival. Women in the richest households and those with higher education are

significantly more likely to deliver in health facilities and attend more ANC visits, compared to their poorer, less-educated counterparts. Notably, an interaction of wealth and education suggests a compounding advantage for women who are both wealthy *and* educated. These findings underscore persistent disparities in healthcare access and highlight the need for targeted interventions to bridge gaps based on socioeconomic background. Policy implications include strengthening and expanding incentive programs (such as Nepal's Aama program for free maternity services) and educational initiatives to ensure equitable maternal healthcare for all groups.

1. Introduction

Maternal and child health remain pressing public health concerns in low- and middle-income countries, and Nepal is no exception. While the country has made significant strides in reducing maternal and infant mortality rates over the past two decades, these improvements have not been equally distributed across socioeconomic groups. Women from poorer households, with less education, and living in rural areas are consistently less likely to access antenatal care (ANC), deliver in health institutions, or receive postnatal checkups. These disparities in maternal healthcare access and utilization persist despite government initiatives like the Aama Program, which provides financial incentives for healthcare visits. The unequal uptake of maternal health services highlights deeper structural constraints related to poverty, education, and geography.

This paper investigates the relationship between household socioeconomic status and maternal healthcare utilization in Nepal. Specifically, I ask: How do wealth and maternal education affect the use of maternal health services and infant health outcomes? I analyze four outcomes: (1) the number of ANC visits, (2) whether the most recent birth took place in a health facility

(institutional delivery), (3) whether the mother received a postnatal checkup within 2 months of delivery, and (4) the number of infant deaths experienced by the respondent. In addition to estimating average effects, I also explore whether wealth and education interact in ways that compound or offset disadvantage.

The analysis draws on the 2016 Nepal Demographic and Health Survey (NDHS), a nationally representative survey of women aged 15–49. I restrict the sample to women who reported a live birth in the five years preceding the survey. The outcome variables include both binary and continuous measures, requiring logistic regression models for institutional delivery and postnatal checkups, and ordinary least squares (OLS) models for ANC visits and infant mortality. The main explanatory variables are household wealth quintile (categorized as poorest, poorer, medium, richer, richest) and maternal education (categorized as none, primary, secondary, and higher), with controls for urban residence, mother's age, number of children. Interaction terms between wealth and education levels are included to test for compounding effects. Robustness checks include provincial fixed effects to account for any regional heterogeneity.

Survey-weighted regression models reveal strong socioeconomic gradients in maternal health service utilization. Women in higher wealth quintiles are significantly more likely to deliver in health institutions and receive postnatal checkups, with the richest women nearly six times more likely to deliver in a facility than the poorest. For ANC visits, education appears to be a more powerful predictor: women with higher education attend up to 1.7 more ANC visits than those with no schooling. While education also improves rates of institutional delivery and postnatal care, the interaction between wealth and education produces super linear gains. Women in the richest quintile with higher education show dramatically higher service utilization rates. Infant

mortality is less responsive to wealth but declines modestly with higher education, particularly when combined with better household economic conditions.

These findings suggest that financial and educational resources not only independently affect maternal healthcare behaviors but also reinforce each other. Nepal's policy focus on cash transfers and cost reduction, while important, may be insufficient on their own to close the equity gap. Interventions that simultaneously improve women's educational attainment and reduce hidden costs—such as lost wages, transportation time, and lack of childcare—could amplify the effects of existing programs. This paper contributes to the literature by incorporating interaction effects and by including postnatal care—an often overlooked but crucial component of the maternal healthcare continuum—in the analysis.

2. Literature Review

Maternal healthcare utilization in Nepal is shaped by a confluence of socioeconomic factors, primarily wealth, education, and geography. Public health studies consistently show that financial resources are critical in overcoming direct barriers to care. For example, analysis based on the NDHS reveal that wealthier households can better afford transportation, fees, and medications, resulting in higher rates of antenatal care (ANC) visits and institutional deliveries (Devkota et al., 2018). Education, on the other hand, plays a dual role: it enhances awareness of health benefits and empowers women to make informed healthcare decisions. Studies by Adhikari (2016) and Sharma et al. (2021) demonstrate that women with higher educational attainment are more likely to seek and utilize maternal health services. Education reduces the influence of detrimental cultural norms and improves women's capacity to navigate the healthcare system.

Economic research reinforces these findings through theoretical and empirical perspectives. Grossman's (1972) health capital model suggests that education increases health production efficiency, leading to improved outcomes. Lleras-Muney (2005) further supports this view by showing a strong association between education and reduced mortality. Additionally, research on income inequality by Black et al. (2013) and Houweling et al. (2007) argues that disparities in wealth are closely linked with unequal access to quality healthcare, disproportionately affecting lower-income groups. Geographical disparities add another layer of complexity. Women in rural areas often face longer distances to health facilities and limited infrastructural support, which, combined with lower income and education levels, further inhibit access to maternal healthcare (Khanal et al., 2019).

Global and comparative studies reinforce the same conclusion: inequalities in maternal care use are widespread. Houweling et al. (2007) document huge poor–rich gaps in maternity care across developing countries. More recent evidence from Ethiopia likewise finds “pro-rich” utilization of ANC, delivery and PNC, driven largely by wealth, education, and urban residence. The Mekonen et al. (2024) study in Ethiopia reports that wealthier women have substantially higher coverage of all essential maternal services (ECI indices of 0.115–0.223 for ANC, delivery, PNC). These findings suggest a common pattern in South Asia and Africa: socioeconomic status powerfully predicts whether a mother obtains adequate care.

By synthesizing insights from both public health and economic research, it is evident that maternal healthcare in Nepal is determined by the cumulative impact of financial, educational, and geographic factors. This integrated approach highlights the need for multifaceted policy interventions that address not only the immediate financial barriers but also the underlying issues

of educational inequality and infrastructural inadequacy, paving the way for improved maternal health outcomes in low-income settings.

Although existing literature robustly identifies individual impacts of wealth, education, and geography, fewer studies have explored interactions among these determinants comprehensively. This paper builds on and contributes to the literature by explicitly examining the interactive effects of socioeconomic status components—particularly wealth and education—on maternal healthcare utilization. Unlike prior analyses that typically assess these variables separately or implicitly, the present study explicitly includes interaction terms between wealth and educational attainment to measure compounding or mitigating effects of combined socioeconomic advantages or disadvantages.

Moreover, the analysis presented in this paper expands the literature by incorporating postnatal care, a vital yet frequently neglected aspect of maternal healthcare, into the examination of socioeconomic gradients (Devkota et al., 2018; Pandey et al., 2020). Existing research largely focuses on antenatal and delivery care, often neglecting the critical postpartum period crucial for maternal and infant health (Singh et al., 2012; Neupane & Doku, 2013). By addressing this gap, this research provides a more comprehensive overview of maternal healthcare utilization patterns.

3. Data Description

This study uses data from the 2016 Nepal Demographic and Health Survey (NDHS), part of the globally coordinated DHS program, designed to provide nationally representative, cross-sectional data on population health, maternal and child health, fertility, family planning, and nutrition (DHS Program, 2017). The NDHS 2016 was conducted between June 2016 and January

2017 and employed a two-stage stratified cluster sampling design, ensuring representativeness across Nepal's seven provinces and ecological zones (mountain, hill, terai).

In the first stage, clusters (enumeration areas) were randomly selected using probability proportional to size sampling from the 2011 National Population and Housing Census frame (CBS, 2014). In the second stage, systematic sampling was used to select 30 households within each cluster, yielding a total sample of 12,862 households. For this analysis, the unit of observation is ever-married women aged 15–49 who reported at least one live birth in the five years preceding the survey, resulting in an analytical sample of approximately 5,060 women.

Data collection was conducted using structured questionnaires administered through Computer-Assisted Personal Interviewing (CAPI) technology by trained enumerators. The survey collected detailed information on maternal healthcare behaviors, socioeconomic characteristics, demographic variables, and child health outcomes. Rigorous quality control procedures, including real-time data transmission, field supervision, and centralized consistency checks, ensured the reliability of the data (DHS Program, 2017).

Key outcome variables in this study include:

- Institutional Delivery: Binary indicator coded 1 if the most recent birth occurred in a formal healthcare facility (hospital, clinic, or health post), and 0 if at home or elsewhere.
- Antenatal Care (ANC) Visits: Continuous variable representing the number of antenatal care visits during the most recent pregnancy.
- Postnatal Checkup: Binary indicator coded 1 if the mother received a postnatal check within two months of delivery.

- Infant Mortality: Continuous variable representing the number of children who died before age one.

Primary explanatory variables include:

- Wealth Quintile: Ordinal categorical variable based on a composite index of household assets, categorized into five quintiles (poorest, poorer, middle, richer, richest).
- Maternal Education: Categorical variable indicating the highest level of education attained: no education, primary, secondary, or higher.

Control variables include:

- Residence (Urban/Rural): Binary indicator coded 1 for urban, 0 for rural.
- Mother's Age: Continuous variable representing the respondent's age at the time of the survey.
- Number of Living Children: Continuous variable indicating parity.

Survey weights provided by NDHS are used in all analyses to account for the complex survey design and ensure nationally representative estimates (DHS Program, 2017).

This study's contribution lies in its detailed examination of socioeconomic disparities using a robust dataset that enables disaggregation by wealth, education, and geographic factors. By applying appropriate statistical weights and accounting for clustering and stratification, the analysis ensures credible, generalizable insights for policy and practice.

Table 1: Summary Statistics

Variable	N	Mean	SD
ANC Visits (m14)	4006	4.2339	2.3028
Institutional Delivery	4857	0.5596	0.4965
Mother's Age (v012)	26 028	35.8145	8.0538
Number of Children (v201)	26 028	3.7900	1.9470

Notes: N is the number of observations; Mean is the sample mean; SD is the sample standard deviation.

Table 1 reports descriptive statistics for our key variables. Among the 4,006 women with at least one recorded ANC visit, the average number of visits is 4.23 (SD = 2.30). Institutional delivery occurs in 55.96 percent of the 4,857 most recent births (SD = 0.50). Looking across the full sample of 26,028 women, the mean age is 35.81 years (SD = 8.05), and respondents report an average of 3.79 children (SD = 1.95). These figures highlight substantial variation in both maternal characteristics and care-seeking behaviors in the NDHS 2016 data.

Table 2: Summary Statistics of Categorical Variables

Panel A: Wealth Quintile

Category	N
Poorest	6,948
Poorer	5,716
Middle	5,378
Richer	4,519
Richest	3,467

Panel C: Urban/Rural

Category	N
Rural	10,519
Urban	15,509

Panel B: Education Level

Category	N
No Education	15,265
Primary	4,584
Secondary	4,775
Higher	1,404

Panel D: Province

Province	N
1	3,276
2	5,153
3	2,725
4	2,812
5	4,039
6	4,029
7	3,994

Notes: Counts are based on NDHS 2016 survey data.

Table 2 summarizes the distribution of categorical variables in the sample, including wealth, education, and residence. The sample is tilted toward lower socioeconomic groups, reflecting Nepal's population structure and possibly differential fertility (poorer and less-educated women

tend to have more children, thus appearing more in our recent-birth sample).

Approximately 27% of observations fall in the *poorest* wealth quintile, compared to only about 13% in the *richest* quintile. In terms of education, the majority of women have low education: nearly 60% have no formal education, around 18% have only primary schooling, ~18% have secondary, and only about 5% have higher education. This stark distribution underscores that a typical mother in the sample is likely to have little or no schooling. The urban–rural split shows about 59% urban and 41% rural in the (weighted) sample of recent births, which suggests that urban areas (where fertility rates are somewhat lower) still contribute a large share of births likely due to their population size and perhaps higher coverage in the survey weighting. Finally, the seven provinces contribute between 10% to 20% each of the sample’s observations. Province 2 (Madhesh) and Province 5 (Lumbini) have slightly larger shares of the country’s recent births in the data (around 19–20% each), whereas Province 3 (Bagmati, containing Kathmandu) has a smaller share of recent births (reflecting lower fertility there). These patterns will be important to keep in mind when we later introduce province fixed effects and discuss regional results.

4. Methods Description:

This study employs both logistic and ordinary least squares (OLS) regression models to estimate the effects of socioeconomic status on maternal healthcare utilization and infant outcomes in Nepal, using survey-weighted data from the NDHS 2016. All models incorporate sampling weights and adjust standard errors for clustering at the primary sampling unit (PSU) level to account for the complex survey design.

Model 1: Logistic Regression for Binary Outcomes

The first set of models focuses on binary dependent variables, specifically:

- **Institutional Delivery** (1 = delivered in a health facility, 0 = otherwise)
- **Postnatal Checkup** (1 = received postnatal care within two months, 0 = otherwise)

$$\log\left(\frac{p_{(ID_i=1)}}{1-p_{(ID_i=1)}}\right) = \beta_0 + \beta_1 WealthQuintile_i + \beta_2 Education_i + \beta_3 Urban_i + \beta_4 Age_i + \beta_5 NumberofChildren_i + \varepsilon_i$$

$$\log\left(\frac{p_{(PNC_i=1)}}{1-p_{(PNC_i=1)}}\right) = \delta_0 + \delta_1 WealthQuintile_i + \delta_2 Education_i + \delta_3 Urban_i + \delta_4 Age_i + \delta_5 NumberofChildren_i + \varepsilon_i$$

where p_i is the probability of the outcome for individual i , $wealthquintile_i$ denotes household wealth quintile dummies, $education_i$ denotes maternal education dummies, and other includes control variables: urban residence, mother's age, and number of living children.

Model 2: OLS Regression for Continuous Outcomes

The second set of models applies OLS regression to continuous dependent variables:

- **Number of Antenatal Care (ANC) Visits**
- **Number of Infant Deaths**

$$ANCVisits_i = \phi_0 + \phi_1 WealthQuintile_i + \phi_2 Education_i + \phi_3 Urban_i + \phi_4 Age_i + \phi_5 NumberofChildren_i + \varepsilon_i$$

$$\text{Numberofinfantdeaths}_i = \alpha_0 + \alpha_1 \text{WealthQuintile}_i + \alpha_2 \text{Education}_i + \alpha_3 \text{Urban}_i + \alpha_4 \text{Age}_i + \alpha_5 \text{NumberofChildren}_i + \varepsilon_i$$

where ANCVisits_i or $\text{Numberofinfantdeaths}_i$ are the continuous outcome for individual i , and the explanatory variables are as defined above.

We estimate a baseline model including wealth and education main effects and controls. Then, to address our hypothesis about interaction effects, we extend the model to include interaction terms between wealth and education levels (e.g., an indicator for being in the richest quintile *and* having secondary education, etc.). This allows us to test whether the effect of, say, being rich is different depending on one's education level (and vice versa). A positive interaction term would indicate a compounded advantage – that high wealth and high education together yield higher utilization than the sum of their separate effects.

Finally, we run a robustness specification adding province fixed effects – i.e., including dummy variables for Provinces 1 through 7. This controls for any time-invariant differences across provinces (such as overall development level, predominant culture, or state of healthcare facilities) that might confound the relationship between SES and healthcare utilization. For example, if poorer provinces also had worse health infrastructure, not accounting for province effects might overstate the wealth coefficient. By including province dummies, we ensure the wealth and education effects are identified from *within-province* variation (comparing women of different SES in the same province).

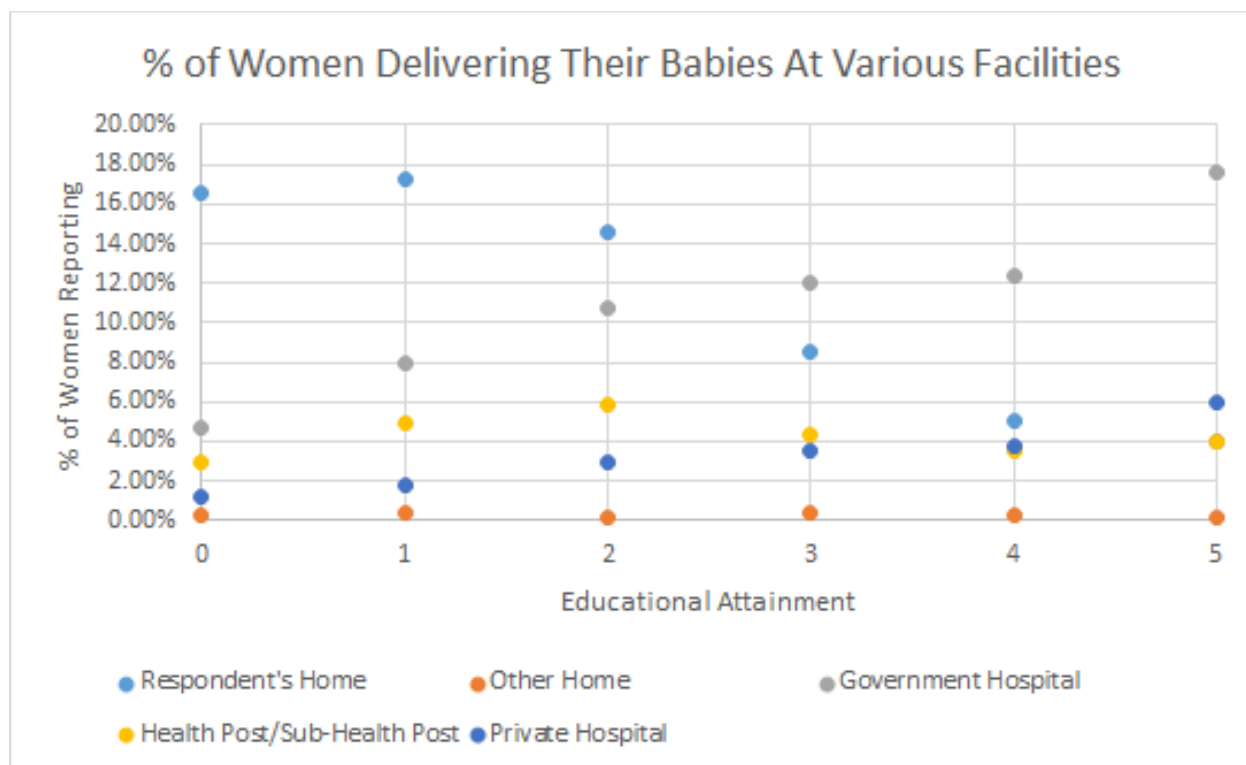
All regressions are weighted using DHS sample weights. We report robust standard errors, clustered at the primary sampling unit (PSU) level, to account for the survey design and any

intra-cluster correlation. Statistical significance is evaluated at the 10%, 5%, and 1% levels, and we denote these with (*) for $p < 0.10$, ** for $p < 0.05$, and *** for $p < 0.01$ in tables.

5. Results

An analysis by educational attainment on the percentage coverage of women delivering their babies at various facilities depicted in figure 1 shows the variation in access to health facilities led by education level. With increase in educational attainment from 0 to 5, the percentage of women delivering their babies at government hospitals increased from 4.69% to 17.57% and private hospitals increased from 1.17% to 5.95%. Women with educational attainment level 1 have the highest percentage of deliveries at respondent's homes (17.30%) and other homes (0.44%) while women with educational attainment level 2 have the highest percentage of deliveries at health posts/sub health posts (5.82%). This shows how access to different health facilities is varied with different level of educational attainment.

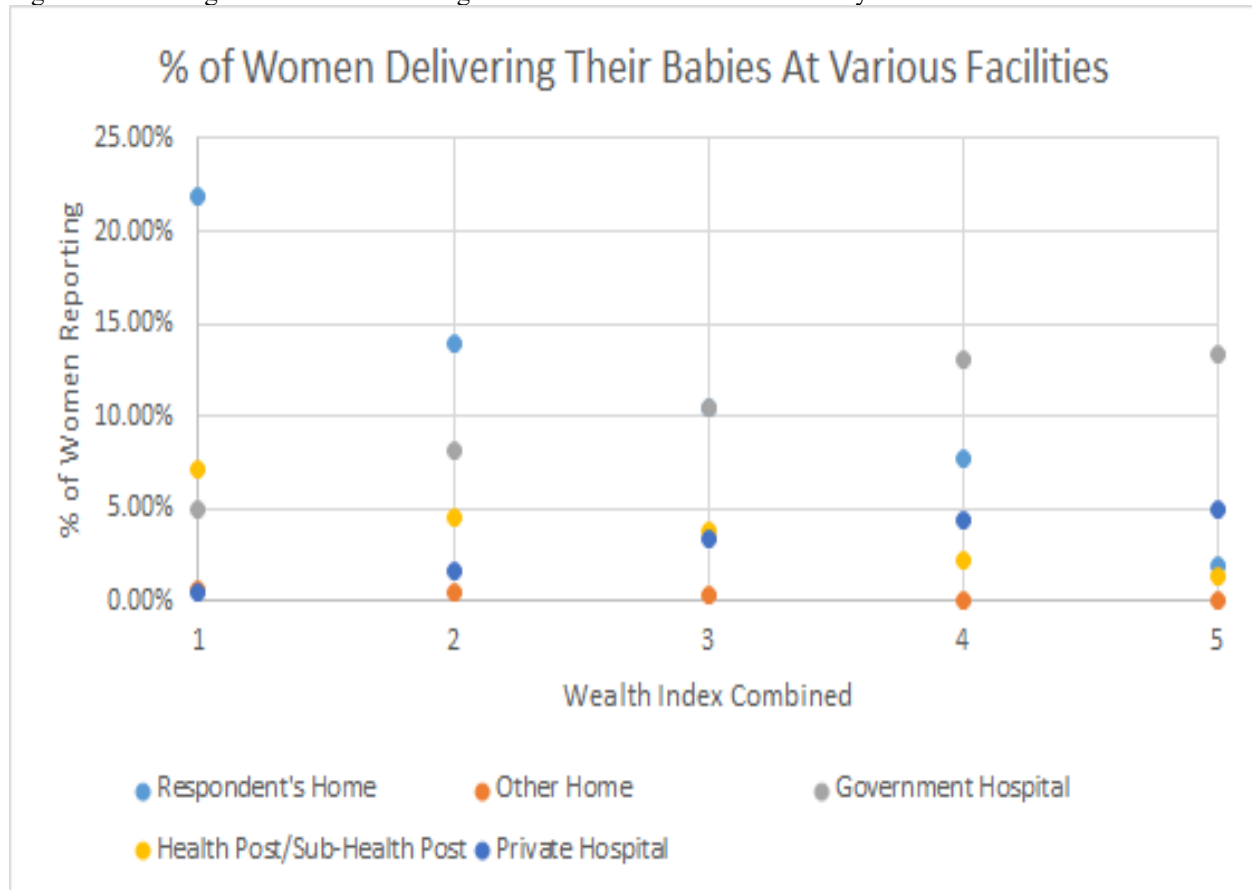
Figure 1: Percentage of Women Delivering Their Babies at Various Facilities by Educational Attainment



Notes: Data are drawn from NDHS 2016. Percentages reflect the proportion of women reporting their most recent delivery location by educational attainment (coded 0–5). Categories include respondent’s home, other home, health post/sub-health post, private hospital, and government hospital. Source: Author’s calculations.

In figure 2, by wealth index, a specific pattern on percentage of deliveries at various facilities was observed: as the wealth index increased from 1 to 5, the percentage of women delivering at respondent's home, other home and health post/sub health posts was decreasing while the rate was increasing for government and private hospitals. Women from poorest wealth index level 1 have the highest percentage of deliveries at respondent's homes (21.85%), other homes (0.59%) and health posts/sub health posts (7.09%) and have the lowest percentage of deliveries at government hospitals (5.03%) and private hospitals (0.44%). The share of delivery in government hospital increases by wealth index which suggests public infrastructure like government hospitals are dependable for delivery for women of all wealth quintiles.

Figure 2: Percentage of Women Delivering Their Babies at Various Facilities by Wealth



Notes: Data are from NDHS 2016. Percentages represent the share of women reporting delivery locations across five wealth quintiles (1 = poorest, 5 = richest). Source: Author's calculations.

Table 3 presents the main regression results for the four outcome variables, including wealth and education as predictors along with control variables (urban/rural, age, parity).

Table 3: Comparison of Survey-Weighted Regression Models

	ANC Visits (1) OLS (Linear)	Institutional Delivery (2) Logistic	Postnatal Check (3) Logistic	Infant Deaths (4) OLS (Linear)
Wealth: Poorer	0.056 (0.205)	0.416* (0.240)	-0.155 (0.231)	0.015** (0.008)
Wealth: Middle	0.554*** (0.212)	0.985*** (0.248)	0.134 (0.205)	0.011 (0.009)
Wealth: Richer	0.471* (0.274)	1.082*** (0.251)	0.228 (0.221)	0.016** (0.008)
Wealth: Richest	-0.014 (0.709)	1.751*** (0.514)	-0.191 (0.501)	0.011* (0.006)
Education: Primary	0.277* (0.161)	0.220 (0.239)	-0.096 (0.189)	0.023 (0.017)
Education: Secondary	0.453*** (0.164)	0.488** (0.203)	-0.198 (0.219)	0.057* (0.035)
Education: Higher	0.605** (0.246)	0.887*** (0.314)	0.171 (0.312)	0.017 (0.031)
Urban Residence	0.171 (0.117)	0.551*** (0.135)	-0.015 (0.097)	0.001 (0.005)
Mother's Age	0.037*** (0.011)	0.026** (0.012)	0.003 (0.010)	-0.003*** (0.0005)
Number of Children	-0.291*** (0.038)	-0.412*** (0.048)	-0.115*** (0.043)	0.009*** (0.002)
Province Dummies^a	Included	Included	Included	Included
<i>Interaction Terms^b</i>	Yes	Yes	Yes	Yes
Constant	3.194*** (0.312)	-0.834** (0.347)	-0.470* (0.265)	0.097*** (0.014)
Observations	4,006	4,857	4,001	26,028
Log Likelihood	-9,069.443	-	-	10,779.950
AIC	18,196.89	-	-	-21,501.90

Notes: Robust standard errors in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

^a Province dummies v0242 to v0247 are included but not displayed for brevity.

^b All interaction terms between wealth and education categories are included.

Notes: Robust standard errors are reported in parentheses. Models include controls for maternal age, number of children, urban residence, wealth quintile, education level, and province fixed effects. Significance levels are indicated by $p < 0.1$ (), $p < 0.05$ (), $p < 0.01$ (). All results are weighted using NDHS 2016 survey sampling weights.

Table 3 shows that both wealth and education increase ANC use, though with diminishing returns. Compared to the poorest mothers, those in the middle and richer quintiles have about 0.55 and 0.47 more ANC visits on average (both statistically significant), whereas the richest quintile's effect is smaller and not significant. Education effects are strong: mothers with secondary education attend ~0.45 more visits ($p < 0.01$) and those with higher education ~0.61 more ($p < 0.05$), compared to no education. Other factors matter too: living in an urban area and

older maternal age both raise ANC use, while each additional child (higher parity) reduces ANC visits by about 0.29 ($p < 0.01$), reflecting resource and time constraints on larger families.

Higher wealth and higher education are both strongly significant predictors of delivering in a health facility. Taking the coefficients to odds ratios: women in the richest quintile have 5 to 6 times higher odds of institutional delivery compared to women in the poorest quintile, holding other factors constant. Even the middle quintile shows markedly higher odds (around 3 times higher odds than the poorest). This indicates a steep gradient: the likelihood of giving birth in a facility rises dramatically with each step up the wealth ladder. Education shows a similarly positive effect – women with secondary education have significantly higher odds of institutional delivery than those with no education (for example, secondary education might correspond to about 2–3 times higher odds, and higher education even more). Interestingly, when both wealth and education are included, both remain significant, implying each has an independent effect. In other words, even at the same wealth level, education makes a difference: an uneducated rich woman is less likely to use maternal health services than an educated rich woman (though still more likely than a poor woman). Among control variables, being in an urban area significantly raises the probability of institutional delivery (as expected, due to better physical access to facilities). Parity has a negative coefficient – each additional child slightly reduces the odds of institutional delivery, suggesting women may be less inclined or able to go to a facility for later births. Maternal age has a positive but small effect (older mothers slightly more likely to use facilities, possibly reflecting accumulated experience or generational differences).

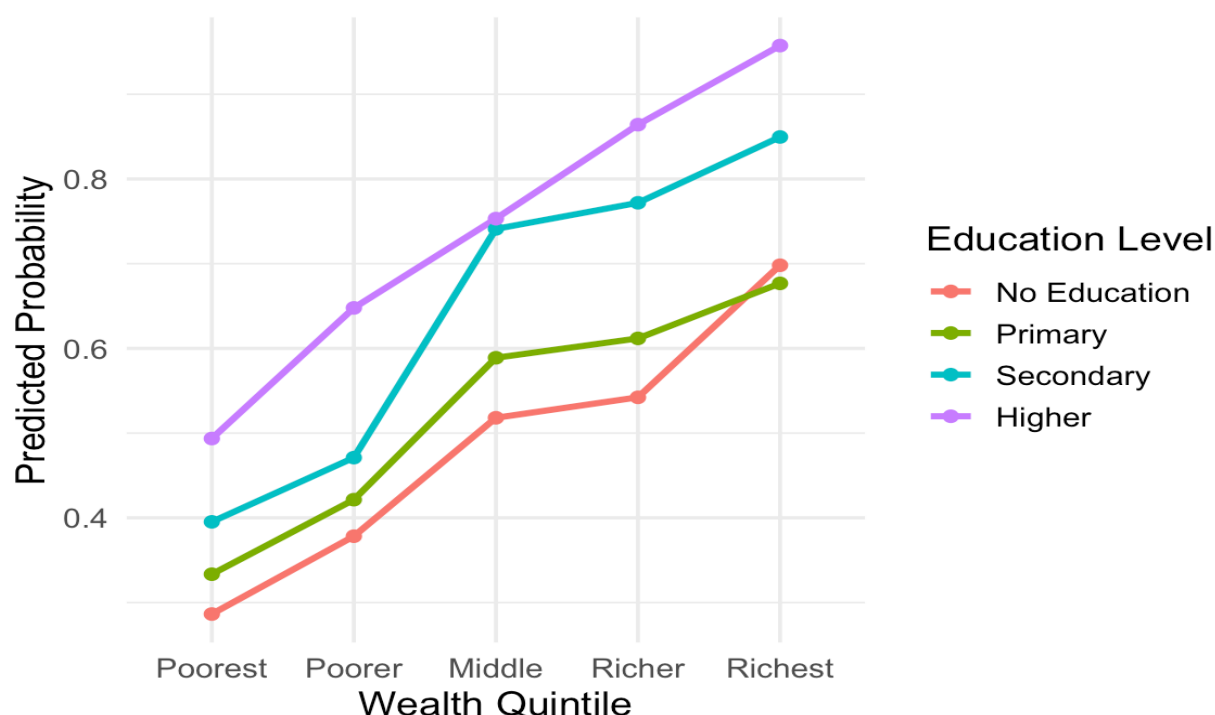
In contrast to ANC and delivery, neither wealth nor education has a statistically significant effect on the probability of receiving a postnatal checkup. All wealth quintile coefficients are small and

insignificant, and the education coefficients are likewise near zero. This suggests that, even though rich and educated women use more ANC and deliver in facilities much more often, these advantages do not translate into higher PNC uptake. The only significant predictor is parity: having more children is associated with a slightly lower probability of postnatal care (approximately -0.12 per child, $p < 0.01$). Overall PNC coverage is low ($\sim 57\%$), and socioeconomic differences appear muted.

Surprisingly, higher wealth alone is not significantly associated with fewer infant deaths in the OLS model. In fact, the richest quintile shows a weakly positive coefficient (not statistically significant), implying no direct wealth advantage after controls. Education has a protective effect: mothers with secondary or higher education tend to report fewer infant deaths (negative coefficients, though marginally significant only in interaction models discussed below). Higher maternal age slightly reduces infant death risk (-0.003 per year, $p < 0.01$), while each additional child greatly increases expected infant deaths ($+0.009$ per child, $p < 0.01$), reflecting that women with many births have a higher chance of at least one loss. These baseline models suggest infant mortality is more responsive to mother's education and family size than to wealth per se.

Figure 3 plots the predicted probability of institutional delivery across wealth quintiles, stratified by education levels. The slope of the lines steepens dramatically for higher education groups, especially among the richest women, demonstrating the interaction effect visually. While wealth increases institutional delivery across all education levels, the gap between no education and higher education widens at higher wealth levels, underscoring the compounding effects of multiple forms of advantage.

Figure 3: Predicted Probability of Institutional Delivery by Wealth and Education



Notes: Predicted probabilities are derived from logistic regression models incorporating interaction terms between wealth quintile and education level, controlling for maternal age, number of children, urban/rural residence, and provincial fixed effects. Source: Author's calculations using NDHS 2016 data.

Table 4 explores how wealth and education jointly affect outcomes, showing provincial fixed effects and wealth $\times$ education interactions. The interactions reveal multiplicative effects of combined socioeconomic status. For ANC visits, the richest mothers with higher education gain about 2.66 additional visits on average relative to the poorest mothers with no education ($p < 0.01$). This is far above the sum of the separate wealth and education effects, indicating a synergistic advantage. Other significant positive interactions (richer $\times$ secondary, richer $\times$ higher) also show that education amplifies the wealth effect. Likewise, for institutional delivery, the interaction of richest $\times$ higher education increases the log-odds by $\sim +1.39$ ($p < 0.05$) relative to

baseline, more than doubling the likelihood of facility birth for that group. In other words, the combination of being both wealthy and well-educated yields dramatically higher utilization than either factor alone.

Table 4: Fixed Effects and Wealth $\times$ Education Interactions Across Models

	ANC Visits (1) OLS (Linear)	Institutional Delivery (2) Logistic	Postnatal Check (3) Logistic	Infant Deaths (4) OLS (Linear)
Province 2	-0.271 (0.186)	-0.495*** (0.187)	-0.246 (0.150)	-0.005 (0.007)
Province 3	0.103 (0.198)	0.215 (0.204)	0.272* (0.164)	-0.003 (0.006)
Province 4	-0.229 (0.184)	0.250 (0.241)	0.106 (0.172)	-0.004 (0.006)
Province 5	-0.306** (0.146)	-0.134 (0.220)	-0.306** (0.154)	0.0005 (0.007)
Province 6	-0.792*** (0.174)	-0.242 (0.241)	-0.438* (0.238)	0.008 (0.010)
Province 7	-0.062 (0.149)	0.916*** (0.255)	0.592*** (0.171)	0.002 (0.009)
Poorer $\times$ Primary	-0.031 (0.270)	-0.040 (0.373)	0.236 (0.309)	-0.035* (0.018)
Middle $\times$ Primary	-0.192 (0.275)	0.067 (0.330)	-0.086 (0.307)	-0.029 (0.019)
Richer $\times$ Primary	0.209 (0.355)	0.065 (0.356)	-0.120 (0.343)	-0.040** (0.018)
Richest $\times$ Primary	1.291 (0.855)	-0.320 (0.591)	0.249 (0.610)	-0.018 (0.019)
Poorer $\times$ Secondary	0.245 (0.233)	-0.107 (0.285)	0.340 (0.320)	-0.058 (0.036)
Middle $\times$ Secondary	0.270 (0.250)	0.491 (0.314)	0.204 (0.291)	-0.071** (0.035)
Richer $\times$ Secondary	0.609* (0.310)	0.562* (0.303)	0.283 (0.288)	-0.072** (0.036)
Richest $\times$ Secondary	0.942 (0.732)	0.405 (0.514)	0.871 (0.529)	-0.061* (0.035)
Poorer $\times$ Higher	0.501 (0.349)	0.219 (0.393)	0.453 (0.408)	-0.045 (0.032)
Middle $\times$ Higher	0.645 (0.419)	0.156 (0.499)	-0.269 (0.404)	-0.043 (0.031)
Richer $\times$ Higher	0.827* (0.429)	0.793 (0.488)	-0.192 (0.442)	-0.020 (0.034)
Richest $\times$ Higher	2.655*** (0.915)	1.388** (0.701)	0.820 (0.585)	-0.030 (0.031)
Observations	4,006	4,857	4,001	26,028
Log Likelihood	-9,069.443	—	—	10,779.950
AIC	18,196.890	—	—	-21,501.900

Notes: Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
Only provincial fixed effects and wealth $\times$ education interaction terms shown.

Notes: Only interaction terms between wealth quintile and education level, as well as provincial fixed effects, are shown. Full model specifications include the same controls as Table 1. Robust standard errors are reported in

parentheses. Significance levels: $p < 0.1$ (), $p < 0.05$ (), $p < 0.01$ (). Source: Author's calculations using NDHS 2016 data.

For infant mortality, some negative interactions emerge, implying protective synergy. For example, the coefficients for poorer×primary education, richer×secondary, and richest×secondary are significantly negative (on the infant deaths outcome). This suggests that even among less wealthy households, higher maternal education can reduce infant death risk. In sum, the interaction models indicate that educational attainment can buffer some of the risks associated with lower wealth, but conversely wealth and education together greatly amplify access to care.

Together, the results paint a clear pattern: socioeconomic status is a powerful determinant of maternal healthcare use in Nepal. Wealth and education each raise service use (especially ANC visits and facility delivery), and their combination produces even larger increases. The exception is postnatal care, where utilization remains uniformly low and unresponsive to SES. Infant survival benefits most from education and smaller family size. These findings underscore that major disparities persist: the most advantaged women (rich and educated) enjoy far better access to maternal services than the poorest and least educated.

6. Conclusion

This study highlights persistent inequities in maternal healthcare in Nepal. Using NDHS 2016 data, we find that higher household wealth and maternal education independently and jointly increase the likelihood of receiving antenatal and delivery care. For instance, women who are both in the highest wealth quintile and have higher education attend many more ANC visits and almost always give birth in facilities, whereas the poorest and uneducated women have very low

utilization rates. Importantly, the interaction analysis reveals that combined advantages are multiplicative: the richest, most educated women have dramatically higher service use than implied by the sum of wealth and education alone. By contrast, postnatal care uptake shows virtually no gradient – suggesting that new mothers face barriers to PNC regardless of socioeconomic status. Infant mortality also varies by education (with more educated mothers reporting fewer infant deaths) but not strongly by wealth.

These results are consistent with a large body of literature showing poor–rich and education-based disparities in maternal care, but they add two key points: first, that wealth and education together amplify access, and second, that PNC is exceptionally neglected. The finding that postnatal care remains low even among privileged groups is troubling, given the critical importance of the postpartum period for maternal and newborn health. It implies that targeted efforts may be needed to encourage PNC for all women, for example by integrating it with immunization or nutritional programs.

There are several limitations. The NDHS is cross-sectional, so we cannot infer causal effects. Unmeasured factors (e.g. local health facility quality, cultural practices, or health knowledge) may confound the associations. Self-reported data may have recall bias, especially for events like infant death. Our measure of infant mortality (number of infant deaths per mother) is crude and depends on parity; mothers who lost a child and then did not report may bias results. We include only surviving mothers, so cases where the mother died in childbirth (and infant mortality) are not captured. Finally, the data are from 2016; healthcare programs in Nepal have evolved (e.g. changes in incentives), so current patterns may differ.

Future research could build on this analysis in several ways. A qualitative study could explore why PNC is so underutilized even by wealthy women – for example, are services unavailable, or do women underestimate their importance? A follow-up DHS (when available) could test whether the same patterns hold after recent policy changes. If panel or administrative data could be linked, one could study causal impacts of education (e.g. reforms) or of cash transfer programs on maternal care. Comparative studies might examine if similar wealth–education interactions exist in other South Asian countries. On the supply side, one could analyze whether increasing health infrastructure in poor areas narrows the gaps. In terms of policy, the results imply that simply providing cash incentives (as in Nepal’s Aama program) may not be enough: efforts to improve female education and health literacy, and to reduce indirect costs (transportation, opportunity cost) for poor women, are likely necessary to achieve more equitable maternal health coverage (Bhatta, 2022).

References:

1. Adhikari, Radha. (2016). “*Effect of Women’s Autonomy on Maternal Health Service Utilization in Nepal: A Cross-Sectional Study.*” *BMC Women’s Health*, 16: 26. <https://doi.org/10.1186/s12905-016-0305-7>.
2. Aryal, Krishna Kumar, Sharad K. Sharma, Mukti N. Khanal, Bihungum Bista, Shiv Lal Sharma, Shambhu Kafle, and Mona M. Steffen. (2019). *Maternal Health Care in Nepal: Trends and Determinants*. DHS Further Analysis Reports No. 118. Rockville, MD: ICF. (Analysis of the 2016 NDHS data).
3. Central Bureau of Statistics (CBS). (2014). *National Population and Housing Census 2011 (National Report)*. Kathmandu, Nepal: CBS. (Provides the sampling frame and demographic data used for NDHS clustering)
4. Devkota, Hem Raj, Elizabeth Murray, Michael Kett, and Nisha Groce. (2018). “Are Maternal Healthcare Services Accessible to Vulnerable Groups? A Study Among Women with Disabilities in Rural Nepal.” *PLOS ONE*, 13(7): e0200370. <https://doi.org/10.1371/journal.pone.0200370>.
5. DHS Program. (2017). *Nepal Demographic and Health Survey 2016 – Final Report*. Kathmandu, Nepal & Rockville, MD: Ministry of Health, Nepal and ICF. (Primary source of data used in this study)
6. Grossman, Michael. (1972). “On the Concept of Health Capital and the Demand for Health.” *Journal of Political Economy*, 80(2): 223–255.
7. Houweling, Tanja A. J., Carine Ronsmans, Oona M. R. Campbell, and Anton E. Kunst. (2007). “Huge Poor–Rich Inequalities in Maternity Care: An International Comparative

Study of Maternity and Childcare in Developing Countries.” *Bulletin of the World Health Organization*, 85(10): 745–754.

8. Lleras-Muney, Anna. (2005). “The Relationship Between Education and Adult Mortality in the United States.” *Review of Economic Studies*, 72(1): 189–221.
9. Mekonen, Asnakew Molla, Natnael Kebede, Atrsaw Dessie, Setegn Mihret, and Yawkal Tsega. (2024). “Wealth Disparities in Maternal Health Service Utilization Among Women of Reproductive Age in Ethiopia: Findings from the 2019 Mini-DHS.” *BMC Health Services Research*, 24: 1034. (Demonstrates similar SES gradients in a comparable context)
10. Sharma, Mohan Kumar, et al. (2021). “Maternal Health Care Services in Nepal: A Qualitative Perspective Based on the Socio-Ecological Model.” *Journal of Health Promotion*, 9(1): 42–54. <https://doi.org/10.3126/jhp.v9i01.40961>. (Explores barriers and facilitators to maternal service utilization qualitatively)
11. Timilsina, Purnima, and Bimala Bhatta. (2022). “Maternal Health Care Service Utilization in Nepal: Trends and Inequalities.” *Journal of Neurosurgery Imaging and Techniques*, 7(1): 460–466. (Analyzes inequality in maternal health utilization; secondary source referenced as Bhatta, 2022 in text).